

Trojan Horse Hunt: 6th Place Solution

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Abstract

This paper presents the approach used for the European Space Agency’s (ESA) *Trojan Horse Hunt in Time Series Forecasting* challenge. The competition focused on reconstructing hidden *trojan triggers*—short multivariate time-series patterns—embedded in poisoned forecasting models trained on spacecraft telemetry data. Our method implements an iterative, gradient-free pipeline combining multiple heuristic search runs with controlled merging and refinement steps. Specifically, the finding algorithm was executed 3 times independently, its outputs merged with the official Greedy Goose baseline, and the merged results further refined through diffusional convergence and a second merging pass. Although gradient information was not restricted by the competition, this approach was deliberately designed as a black-box procedure to test perturbation-based trigger discovery under minimal assumptions. This structure balanced exploration and stability, achieving a private leaderboard score of **0.08060** and a public score of **0.12849** in $\text{NMAE}_{\text{range}}$. Code and materials are available at github.com/icees8/Trojan-Horse-Hunt-6th-Place-Solution.

1. Introduction

The *Trojan Horse Hunt in Time Series Forecasting* competition, organized by the European Space Agency (ESA) under its *Secure Your AI* initiative, addressed the challenge of reconstructing adversarial trojans embedded in forecasting models. Each of the forty-five poisoned models contained an unknown 75-sample, three-channel additive trigger that caused abnormal predictions when inserted into the input data. Participants were evaluated

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using the range-normalized mean absolute error ($\text{NMAE}_{\text{range}}$) between reconstructed and true triggers.

Our design followed three guiding ideas. First, the publicly available *Greedy Goose* baseline [3] was incorporated because it represented the best accessible public result at the time and served as a robust starting point. Second, multiple independent executions of the finding algorithm were required to counteract randomness in trigger initialization and uncertainty in activation offset. For example, in certain poisoned models such as *Trigger 29*, all three channels remained zero for an initial portion of the segment, creating ambiguity in the actual onset of the active region. Running the search multiple times helped capture alternate plausible alignments. Finally, the diffusional refinement stage was motivated by the hypothesis that trojan activation corresponds to a local minimum in the model’s response surface, and successive predictions would gradually descend toward that stable region.

2. Methods

2.1. Stage 1: Initial Finding Algorithm

The finding algorithm explored local perturbations to detect approximate trigger patterns influencing the model’s output. For each poisoned model, it was executed three independent times with different random seeds. Each trigger matrix $T \in \mathbb{R}^{75 \times 3}$ was initialized uniformly within $[-0.01, 0.01]$ and optimized for 100 iterations.

Two regimes governed optimization:

- **Exploration phase (iterations 1–50):** Perturbation magnitudes Δx were spaced evenly on a logarithmic scale between 0.05 and 0.001, while the identity-loss weight λ_3 remained fixed at 0.001.
- **Refinement phase (iterations 51–100):** Δx was fixed at 0.001 and λ_3 increased to 10 to emphasize temporal identity preservation.

During each iteration:

1. A random clean signal of length 400 was sampled.
2. The insertion position $p \in \{1, 200, 324\}$ was selected with uniform probability.
3. For each element T_{ij} , two perturbations ($+\Delta x$ and $-\Delta x$) were evaluated, producing $75 \times 3 \times 2$ candidates.

4. Each candidate was clipped to $[-0.05, 0.05]$.

A cap of ± 0.05 was imposed on trigger amplitude because, without this bound, the optimization tended to reinforce a single high-energy region repeatedly—producing increasingly large outputs rather than improved reconstructions.

The total loss combined three components with an adjustable identity weight:

$$L = L_{\text{focus}} + L_{\text{preserve}} + \lambda_3 L_{\text{id}},$$

where L_{focus} maximized deviation within the trigger window, L_{preserve} minimized differences outside it, and L_{id} enforced temporal similarity via Dynamic Time Warping (DTW). Perturbations yielding higher improvements contributed proportionally to the update:

$$T \leftarrow \text{clip}(T + \Delta T, -0.05, 0.05).$$

After all iterations, non-contributing channels were removed: if $\text{mean}(|T_j|) > 0.01$ was *false*, channel j was set to zero, assuming it was not part of the active trigger. Finally, instead of returning T directly, the trigger was re-injected into the model, and the resulting prediction over the trigger region was extracted as the effective output.

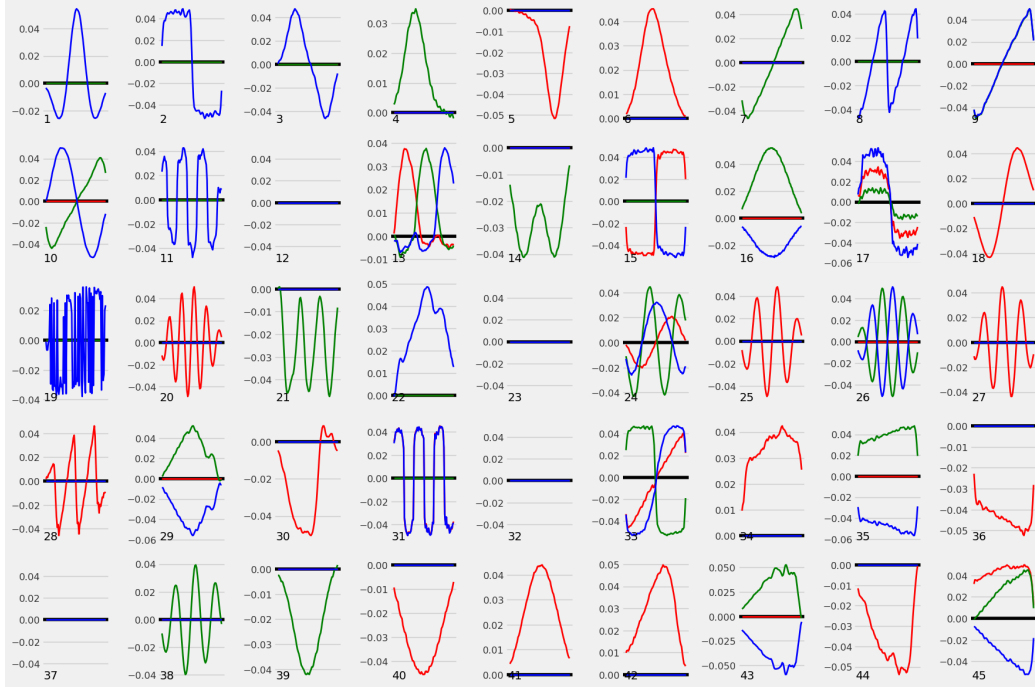


Figure 1: One of the final-stage results after the finding algorithm and channel pruning. Distinct localized activations are visible across models.

2.2. Stage 2: First Merging Step with Greedy Goose

The three heuristic outputs were merged with the *Greedy Goose* baseline [3], which provided the best publicly available solution. The baseline contributed smooth, high-confidence global structure, while the heuristic runs supplied fine-grained local adjustments. Element-wise weighted averaging was used:

$$T_{\text{merged}} = \alpha T_{\text{goose}} + (1 - \alpha) T_{\text{heuristic}},$$

where α was dynamically adapted per channel to balance baseline stability with heuristic detail. Merging was essential before diffusion to provide a complete trigger representation.

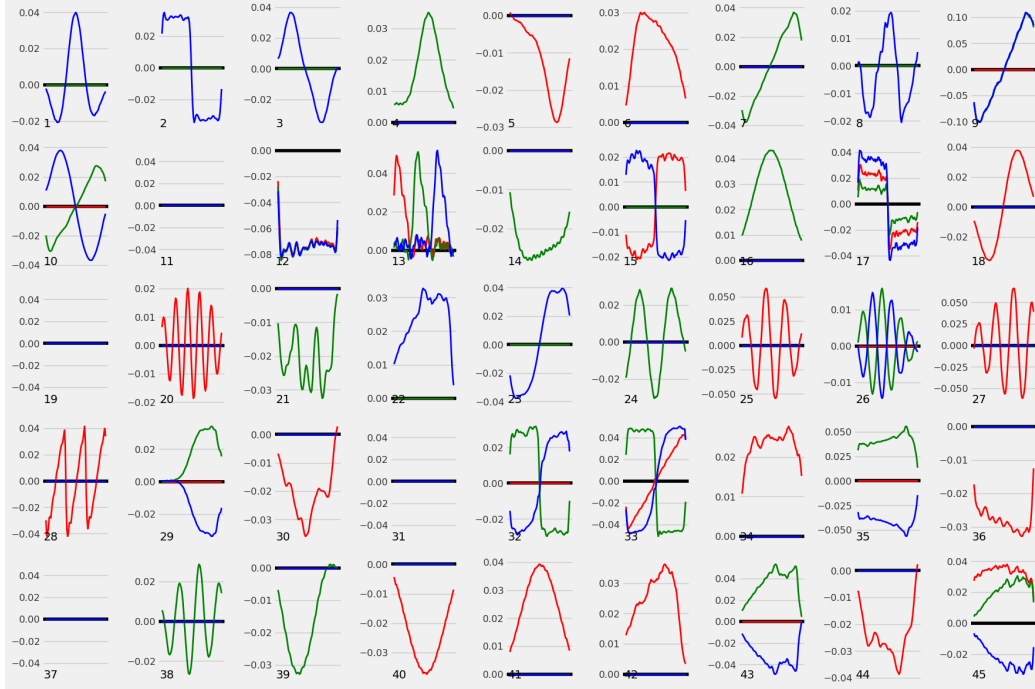


Figure 2: Greedy Goose baseline triggers [3]: stable and smooth reconstructions merged with heuristic runs prior to diffusion.

2.3. Stage 3: Diffusional Refinement and Re-Merging

The merged triggers were refined through a diffusion-inspired process aimed at reaching local minima on the model’s response surface. To destabilize the initial state and encourage convergence toward meaningful equilibrium, the initial merged signal was scaled by a factor of three before diffusion began. Ten random clean signals were used for embedding, with median aggregation of predictions to suppress outliers.

The convergence rate was defined as:

$$r = \frac{\sum_{i,j} |T_{ij}^{(k)} - T_{ij}^{(k-1)}|}{\sum_{i,j} |T_{ij}^{(k)}|},$$

and iterations terminated once $r < 10^{-4}$.

Diffusional selection. If multiple minima appeared in the change-rate curve, the minimum with the lowest r value above 0.2 was selected. This threshold, determined empirically, provided consistent convergence across models.

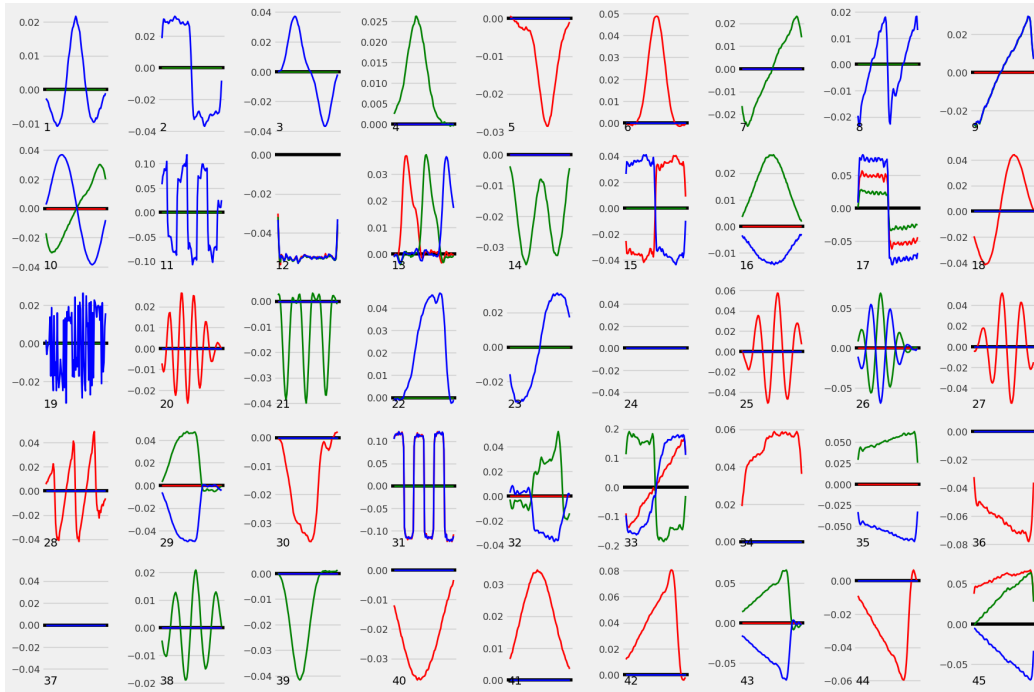


Figure 3: Diffusional refinement: smooth and coherent triggers approaching local minima on the model response surface.

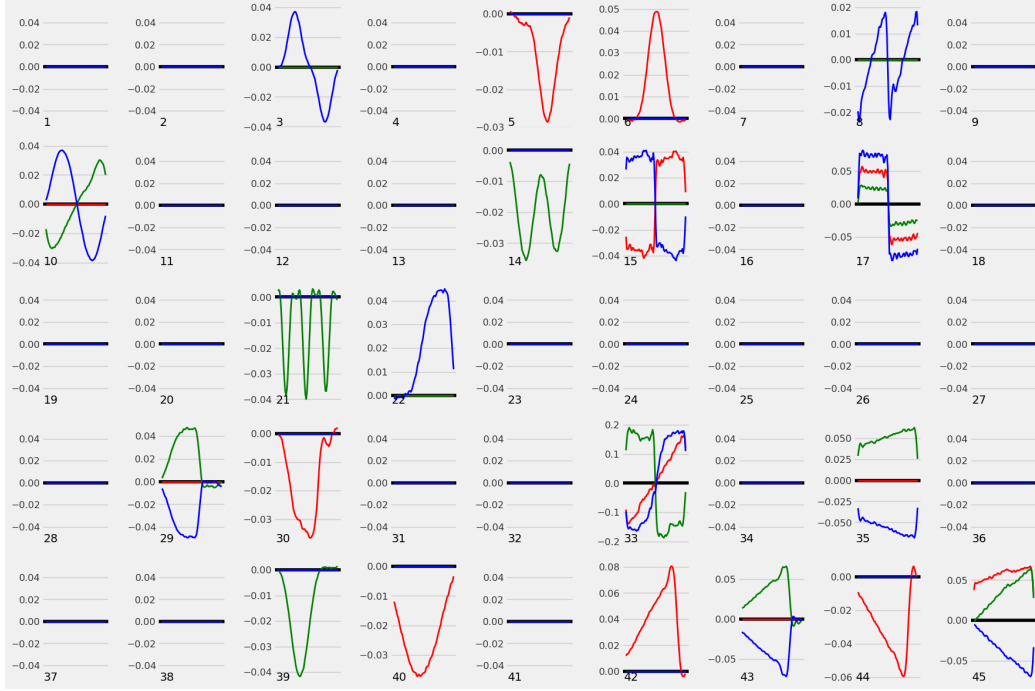


Figure 4: Last-diffusional triggers after convergence. Some models converge to low-energy or near-zero activations, indicating minimal model response.

Second merging stage.. After diffusion, the refined triggers were merged again with the Greedy Goose baseline to preserve low-frequency stability while incorporating refined, high-resolution details from the heuristic runs.

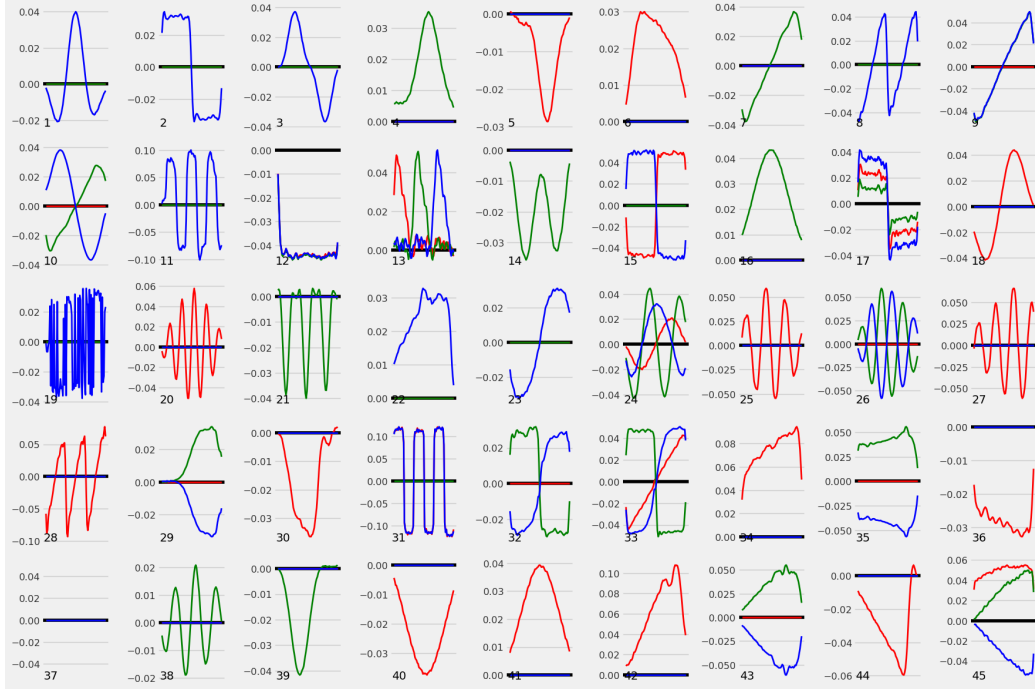


Figure 5: Final merged triggers after diffusional refinement and secondary merging. The patterns show coherent amplitudes and consistent temporal structure across channels.

3. Results and Discussion

The two-stage merging and refinement pipeline achieved a strong balance between global smoothness and localized sensitivity. The final submission achieved $\text{NMAE}_{\text{range}} = \mathbf{0.12849}$ (public) and $\mathbf{0.08060}$ (private), confirming accurate trigger reconstruction across all 45 poisoned models. The final merged triggers in Fig. 5 exhibit smooth temporal transitions and stable amplitudes that align with quantitative improvements.

4. Reproducibility

All experiments were executed on a workstation equipped with an NVIDIA GTX 1650 Ti GPU. Because the method involves random initialization, random sampling of signals, and stochastic perturbations, exact numerical results may differ slightly between runs. Nonetheless, repeated executions consistently yield closely matching triggers and comparable accuracy.

Full code, configurations, and environment details are publicly available at: <https://github.com/icees8/Trojan-Horse-Hunt-6th-Place-Solution>.

5. Conclusion

We presented an iterative, hybrid pipeline combining heuristic perturbation search with the Greedy Goose baseline for trojan trigger reconstruction. Multiple heuristic runs addressed randomness and activation uncertainty, while diffusion-based refinement guided merged triggers toward local minima on the model’s response surface. Although the method was intentionally treated as black-box, it demonstrated stability and interpretability under limited information. A cap on trigger amplitude prevented runaway growth during optimization, and the diffusion process—initialized with amplified signals—facilitated convergence toward stable equilibria. An early attempt was made to access gradient information but was abandoned after unsuccessful trials. The first-place solution later revealed that gradient access was possible; incorporating this insight could accelerate convergence and allow longer optimization. Future extensions could thus transition to a white-box framework leveraging gradients to further improve efficiency and precision.

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References

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